

3.1.3 Scanning

We can consider footprinting the equivalent of taking images of a bank safe and finding out its make and model, and noting the routes of the bank guards, finding out who has the key. Scanning is more intrusive, and involves actually interacting with the target systems. In this step we would follow the “footsteps” and do some actual testing.

Now that we have an idea of the scope of the network that we need to infiltrate, we start poking and prodding at the computers on the network looking for systems that are vulnerable. Specifically, we look for systems that are “alive”, i.e. listening for input on some or the other port.

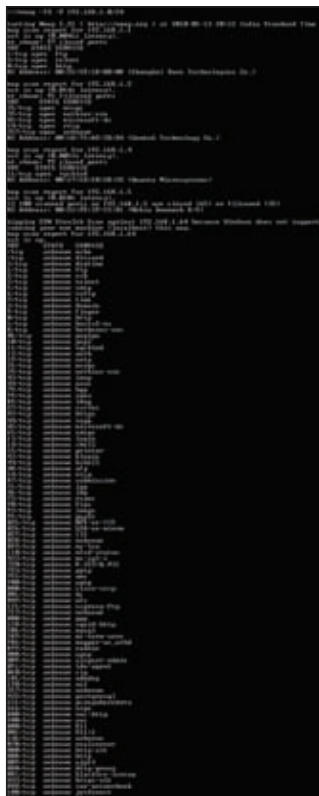
By using a mass ping utility, one can get an idea of which systems on the network are alive, so that we can further test for security lapses.

A more comprehensive look at the systems on a network can be obtained by using utilities such as “Nmap”. This handy tool is a command-line utility that can do everything from a simple ping sweep to a fully comprehensive scan of all open ports on a system.

ZeNmap is an easy to use GUI frontend for Nmap that lets you browse the information provided by Nmap in a much richer way. It includes common network testing profiles that define how much is scanned. You can provide it an IP range to scan and it can construct a topological map of all connected systems in the network.

A quick Nmap scan will reveal all open ports in the system, and as far as possible, it will also reveal the services running on the port. With ident scanning, Nmap will also reveal the owner of each running service – as in the privileges the service is running with. Knowing the identity of the user can help with the attack, as by attacking a service which is running with a privileged (i.e. root) account we stand to gain full root access.

We have used Nmap exclusively in our examples till now, however this does not mean it is the only tool, or even the only good tool. Strobe, Netcat, SuperScan, and many more applications are available which can perform



nmap running in the Windows console, scanning all IPs in the local network